

The **GOVERNMENT SUPPORT** Needs To **EXTEND** For Wider Adoption Of **ROBOTS IN INDIA**

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Imagine an organisation devoid of threat from automation, where humans and robots complement each other on the workflow and take productivity to the next level! The result is an exponentially profitable business! Pradeep David, general manager - South Asia, Universal Robots, in an interaction with Baishakhi Dutta, assures that this is no more a dream

Q. How are collaborative robots (cobots) different from other robots/automation processes?

A. A cobot is designed to work with humans in the same physical work-space without any protective fencing—a feat achieved using external sensory technology. It stops when it encounters anything in its path, unlike an industrial robot that is known to injure and even kill humans in extreme cases.

A cobot is light, small, mobile and energy-efficient, in addition to being re-deployable at multiple locations without any changes to the production layout. It can also be put to work within minutes of reaching the worksite.

It requires very little programming to start work, and users do not require prior programming expertise for its deployment—a traditional robot can take weeks to set up before it is operational.

Q. How can cobots help in increasing productivity?

A. Cobots work alongside humans in small-spaced assembly lines without any fencing (subject to application risk assessment).

These can be redeployed time and again for different applications and in different locations. These give an ROI within the first few years.

Footprint of cobots is 190mm (diameter), and these can be floor-, machine- or overhead-mounted.

Q. Are cobots a good place to start for Indian SMEs in their automation journey?

A. Yes, small footprint, low energy consumption and easy programming give SMEs the flexibility of automating different production lines at different times for customised batch production! With none of the traditional costs associated with expert robotic programming, set up and fenced-off work areas,

average payback of cobots have, in some cases, been less than six months; though, on average, it is about two years.

Q. Is it challenging to convince SMEs to move to automation?

A. Surprisingly, no! SMEs supply to MNCs and end-user functions no longer accept variability. End-users want good quality and consistent productivity.

Q. How can cobots help in electronics manufacturing?

A. Electronics manufacturing involves a lot of assembly work, such as picking up the PCB and inserting it into the server of a computer for testing. Cobots can easily do this.

Another example is inspection. A PCB has thousands of capacitors and tiny electronic components. Cobots can hold a camera and quickly tell whether something is missing.

Q. Any successful deployments in the Indian electronics industry?

A. Yes, we do, but we are not at liberty to reveal any names due to non-disclosure agreements.

Q. Any training required to handle cobots on the work floor?

A. When a company buys a cobot, we train them on how to operate and repair it. Any person working with the cobot becomes a programmer and can train the cobot. No specialised training is required.

Q. What about standardisation regarding implementation of cobots?

A. ISOCS16066 is a global standard that is ratified over the world, and it governs the safety of a human being working with a robot.

Our category of robots are force-limited robots, which means the amount of

force these can apply on a human being is limited. Hence, these cannot crush or kill the human.

Q. How does the Indian market for robotics compare with other countries?

A. India lags in robot adoption with only three robots per 10,000 employees as compared to the global average of 74. Even though the installed base is small, market potential remains huge.

Q. How has your India business fared?

A. In 2018, we hit a revenue of US\$ 234 million. Operations in India commenced in 2016 and have been growing rapidly since then with acceptance from both MNCs and SMEs. Earlier, our customers in India were being served by Universal Robots.

Q. What is your business strategy?

A. We only go to markets through our partners—85 to 90 per cent of our business is through them.

Q. How do you plan to expand in India?

A. We continue to invest in channel partners that can provide domain-specific solutions to end-users. We have a strong distribution network across India, with over fifteen distributors with varying capabilities.

We are looking to expand our network in India through academia, too. Universal Robots Academy—a new online self-training programme—helps acquire skills to program and operate our cobots without further assistance.

Q. Have the government policies had any impact on your business here?

A. Government policies have not been favourable, as there are still import duties on robots in India. The government support needs to extend for wider adoption of robots in India. **EFY**